POPESCU'S THEOREM: EVERY REGULAR HOMOMORPHISM OF NOETHERIAN RINGS IS A FILTERED COLIMIT OF SMOOTH RING MAPS

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Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Reduction to the geometrically regular over-a-field case. lemma-reduce-to-field (carded L26811). Reduces the theorem to k -> Lambda with Lambda Noetherian and geometrically regular over k.
- Solving when the singular ideal generates the unit ideal. lemma-final-solve (carded L26797). Converts h_B = Lambda into the smooth factorization A -> B -> Lambda.
- Resolution in the separable residue case. lemma-resolve-special (carded L26816). Resolves k -> A -> Lambda superset q when the characteristic is zero.
- General resolution in positive characteristic. lemma-resolve-general (carded L26818). Resolves k -> A -> Lambda superset q when the characteristic is p > 0.

Source reference. Original lemma 07EU: sources/stacks-smoothing/pinned-original-source.tex, source lines 529–529. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07F5: sources/stacks-smoothing/pinned-original-source.tex, source lines 2220–2220. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FJ: sources/stacks-smoothing/pinned-original-source.tex, source lines 2889–2889. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FE: sources/stacks-smoothing/pinned-original-source.tex, source lines 2569–2569. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 07F7: sources/stacks-smoothing/pinned-original-source.tex, source lines 2296–2296. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Theorem 0.1 (Popescu)**.** *Any regular homomorphism of Noetherian rings is a filtered colimit of smooth ring maps.*

Proof. By Lemma 07F5 it suffices to prove this for $k \to \Lambda$ where Λ is Noetherian and geometrically regular over k. Let $k \to A \to \Lambda$ be a factorization with A a finite type k-algebra. It suffices to construct a factorization $A \to B \to \Lambda$ with B of finite type such that $\mathfrak{h}_B = \Lambda$, see Lemma 07EU. Hence we may perform Noetherian induction on the ideal $\mathfrak{h}_A$. Pick a prime $\mathfrak{q} \supset \mathfrak{h}_A$ such that $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$. It now suffices to resolve $k \to A \to \Lambda \supset \mathfrak{q}$ (as defined in the text following Situation 07F7). If the characteristic of k is zero, this follows from Lemma 07FE. If the characteristic of k is $p > 0$, this follows from Lemma 07FJ. □